POUGHKEEPSIE PDL
Dept. 539, Bldng. 965
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FILE MEMORANDUM

SUBJECT: Gap-less Information in Real Time

BY: W. Buchholz

In some real time applications there is a need to be able to read information continuously for long periods of time without gaps in the data. The multiple mode provides a means for creating endless "loops" in memory in which the information is stored for processing.

The Chain Flag and Multiple Flag in the CW are set to 1. Whenever the count goes to zero, the Exchange sends a "disconnect" signal to the I/O control unit. The unit immediately returns an "end of message" signal which causes the Exchange to start again with a new CW. (This simulates an inter-block gap even where there is none.) The new CW refers to the new memory area. This device serves to close the loop. COPY CW instructions must be used to determine when the unit has reached the new memory area and processing can begin on the old area.

Chaining in the single mode (Multiple Flag = 0) is not satisfactory because of the requirement for dummy words. It may be difficult to provide for dummy words in real time input.

This mode of operation is different from the usual one where the I/O unit is expected to stop if the computer does not get back to it in time. Here it is necessary to slave the computer to the input and write the program so as to make sure that it gets back in time.

When working in this gap-less mode it is necessary to specify twice the weight count for a given peak information rate to give the Exchange time (equal to one word time) to obtain a new Control Word.

WB/pkb

cc: 7000 PD Planning Group
7000 Product Planning Group
J. D. Calvert
S. W. Dunwell
R. P. Fletcher
J. C. Smith
P. H. Sterbens

H. G. Kolsky